Symmetries, tight binding band structure theory, and effective models

Today’s plan

- Using symmetries to simplify problems
- Single and multiband 1D band-structures
- Single band high dimensional band-structures
- Key properties of electronic dispersions
- ARPES

A reminder from session #1

Hamiltonians can be described in a second quantized formalism

$$H = \sum_{ij} t_{ij} c_i^\dagger c_j$$

Creation operator

Annihilation operator

And can be diagonalized

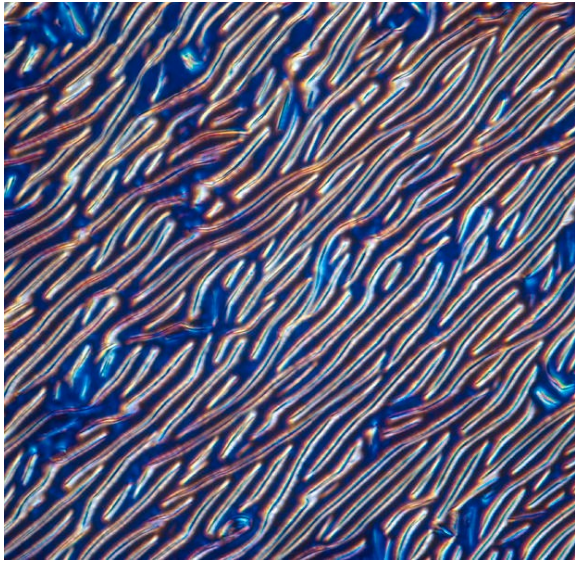
$$\Psi_n^\dagger = \sum_i a_{n,i} c_i^\dagger$$

$$H = \sum_n \epsilon_n \Psi_n^\dagger \Psi_n$$

Eigenenergy

Complexity in nature

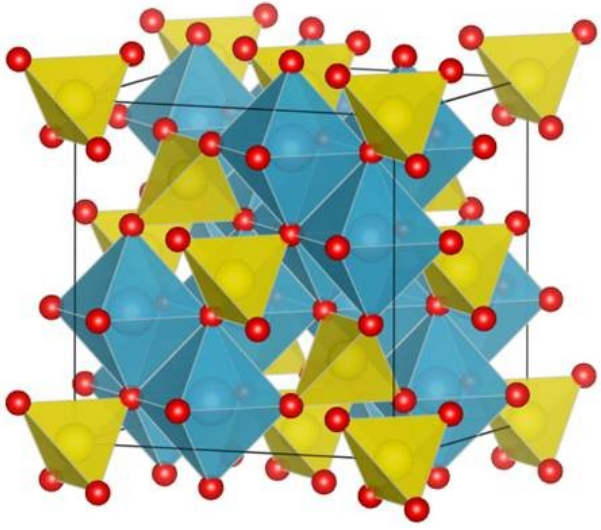
Liquid crystals



Galaxies



Complex materials



How can we make predictions in highly complex systems?

The key idea of symmetry

Symmetries allow to “guess” solutions without explicitly solving a problem

$$f(x) = -f(-x) \quad \rightarrow \quad \int_{-\infty}^{+\infty} f(x)dx = 0$$

$$\left[-\frac{\partial^2}{\partial x^2} + \cos(\Omega x)\right] \Psi(x) = E\Psi(x) \rightarrow |\Psi(x)| = |\Psi(x + 2\pi/\Omega)|$$

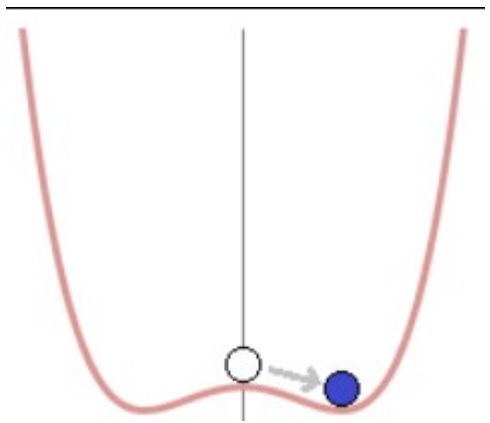
even without understanding the microscopic mechanism governing the system

The key idea of symmetry

The underlying laws of physics are symmetric
But the real system may or may not be symmetric

Non symmetric solution

Spontaneous symmetry breaking



Symmetric solution

Mathematical constrained solution



The role of symmetry

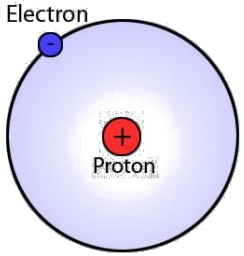
Symmetries enforce observables to vanish

$$\langle A \rangle = 0$$

Certain operator

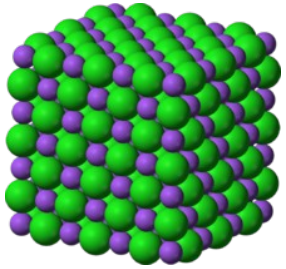
Magnetic moment
Electric dipole
Hall conductivity

Symmetries constrain the mathematical solution of a problem



Hydrogen atom

$$\Psi(r,\theta,\phi) = Y_{lm}(\theta,\phi)R(r)$$



Periodic crystals

$$\Psi(\mathbf{R}+\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{R}}\Psi(\mathbf{r})$$

Why is symmetry important?

Symmetries help to characterize complex problems capturing their physics

New quantum excitations when symmetries are broken (Goldstone modes)

$$H = -\sum_{\langle ij \rangle} \vec{S}_i \cdot \vec{S}_j \longrightarrow \mathcal{H} = \sum_k \epsilon_k b_k^\dagger b_k$$

Symmetry breaking (ferromagnetism)

Interacting local magnetic moments

Emergent bosonic excitations (magnons)

Symmetries allow to greatly simplify quantum problems

Infinite systems can be "folded" to small systems (Bloch's theorem)

$$H = \sum_k H_k$$

"Infinite system" Hamiltonian

(finite) Bloch's Hamiltonian

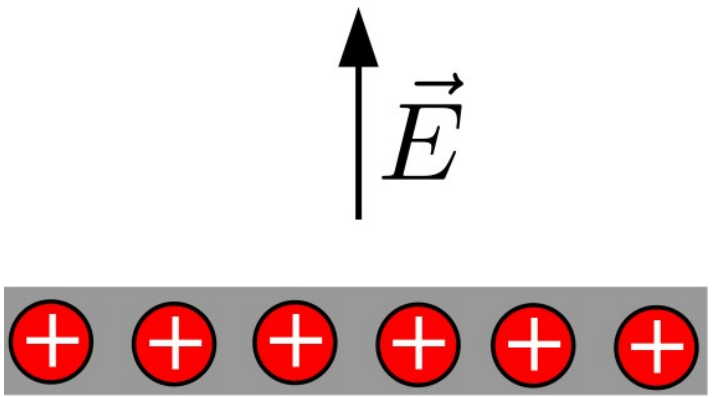
Simplifying problems using symmetries

A familiar example using symmetries

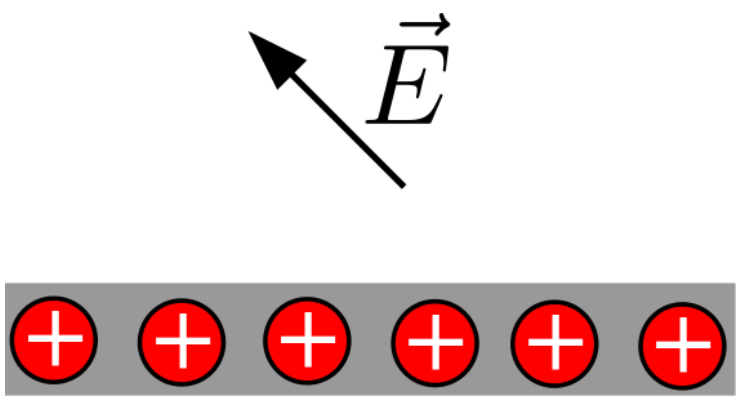
Back to electromagnetism

Imagine an infinite charged plane

Option #1



Option #2



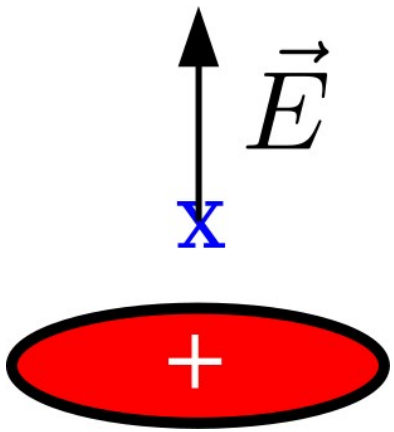
What is the direction of the electric field?

A familiar example using symmetries

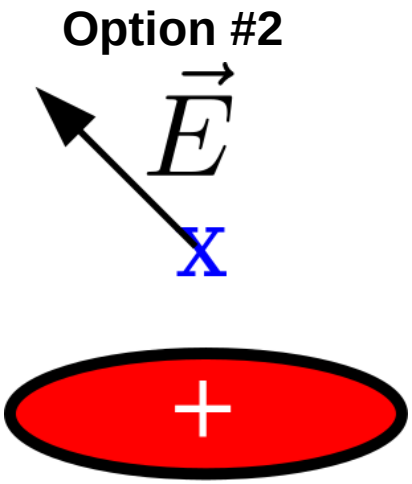
Back to electromagnetism

Imagine an egg-shaped charge

Option #1



Option #2

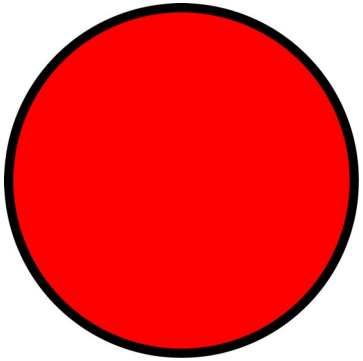


What is the direction of the electric field in position x?

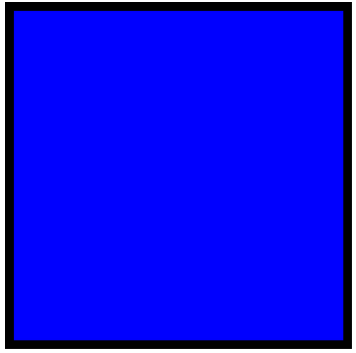
The intuitive notion of symmetry

Symmetry: Transformation performed in a system, that keeps it invariant

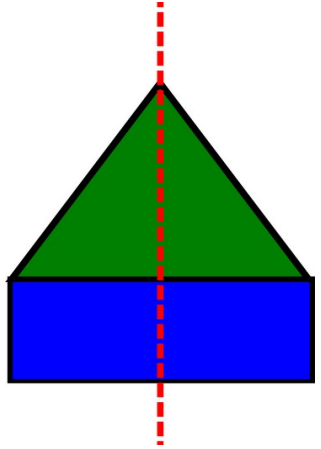
Arbitrary rotation



90 degrees rotation



Right-left reflection



(Symmetry) transformations in quantum physics

How does a transformation affect a wavefunction?

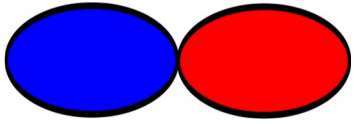
$$|\phi\rangle = S|\Psi\rangle$$

"New" wavefunction

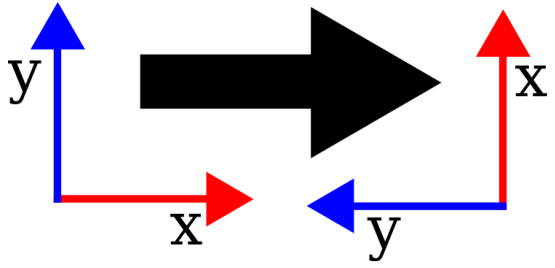
"Old" wavefunction

(Symmetry) transformation

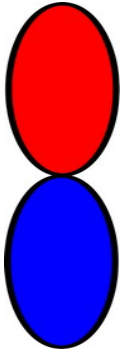
$|\Psi\rangle$



Rotate 90 degrees



$|\phi\rangle$



(Symmetry) transformations in quantum physics

A generic symmetry transformation

$$|\phi\rangle = S|\Psi\rangle$$

By definition, any symmetry transformation must leave any wavefunction normalized

$$\langle\phi|\phi\rangle = \langle\Psi|S^\dagger S|\Psi\rangle = \langle\Psi|\Psi\rangle \longrightarrow S^\dagger = S^{-1}$$

The Hermitian of a symmetry is its own inverse

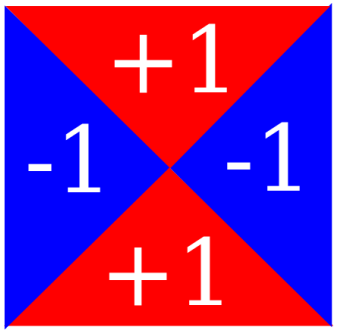
Symmetry transformations

A wavefunction is symmetric under a transformation if $S|\Psi\rangle = \lambda|\Psi\rangle$

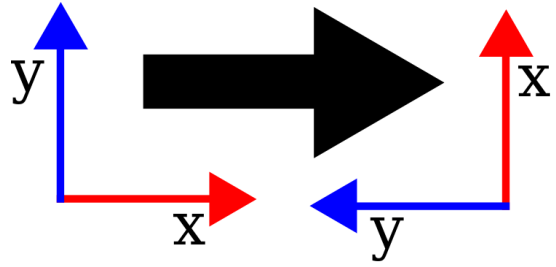
λ ***is the eigenvalue of the transformation***

What is the eigenvalue of this transformation?

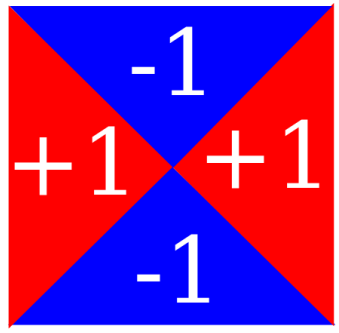
$|\phi\rangle$



Rotate 90 degrees

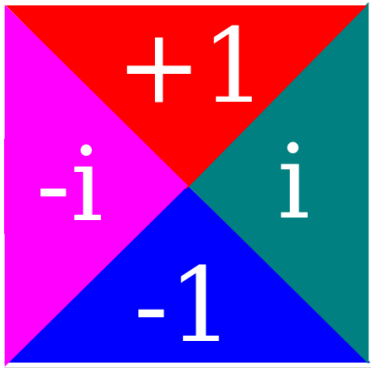


$|\Psi\rangle$

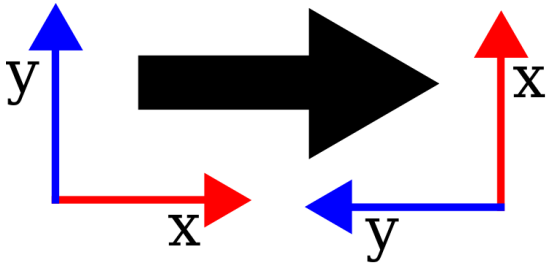


A symmetry transformation

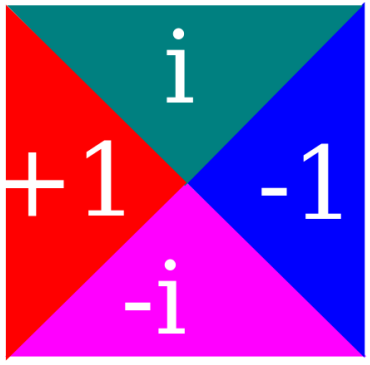
$|\Psi\rangle$



Rotate 90 degrees



$S|\Psi\rangle$

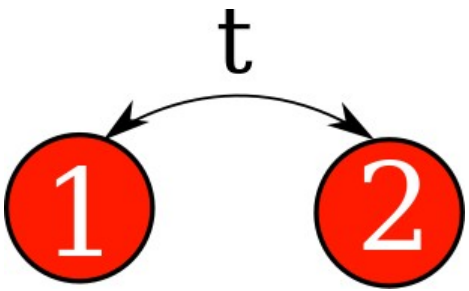


What is the eigenvalue λ of this transformation?

$$S|\Psi\rangle = \lambda|\Psi\rangle$$

Symmetry transformation in a wavefunction

Hamiltonian



$$H = t[c_1^\dagger c_2 + c_2^\dagger c_1]$$

Eigenfunctions $H = t\Psi_\alpha^\dagger \Psi_\alpha - t\Psi_\beta^\dagger \Psi_\beta$

$$\Psi_\alpha^\dagger = \frac{1}{\sqrt{2}}[c_1^\dagger + c_2^\dagger] \qquad \Psi_\beta^\dagger = \frac{1}{\sqrt{2}}[c_1^\dagger - c_2^\dagger]$$

What are their eigenvalues λ_γ under mirror symmetry?

$$c_1 \rightarrow c_2$$
$$c_2 \rightarrow c_1$$

$$S|\Psi_\gamma\rangle = \lambda_\gamma|\Psi_\gamma\rangle$$

Symmetries in a Hamiltonian

A Hamiltonian is invariant under a transformation if

$$SHS^{-1} = H$$

$$[S, H] = SH - HS = 0$$

S Symmetry transformation

H Hamiltonian

A reminder from linear algebra

If two linear operators commute, there is a common basis of eigenstates

$$H|\Psi_n\rangle = E_n|\Psi_n\rangle$$

Hamiltonian eigenvalue

$$S|\Psi_n\rangle = \lambda_n|\Psi_n\rangle$$

Symmetry eigenvalue

Eigenvalues of symmetry operations

Recall the property of symmetry operations

$$S^{\dagger} = S^{-1}$$

All the eigenvalues of symmetry operations are complex numbers in unit circle

$$S|\Psi\rangle = \lambda|\Psi\rangle$$

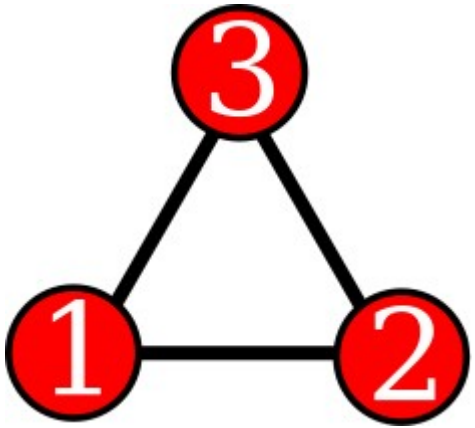
Complex eigenvalue

Real number

$$\lambda = e^{i\phi}$$

Guessing the form of wavefunctions

Take this Hamiltonian



$$H = c_1^\dagger c_2 + c_2^\dagger c_3 + c_3^\dagger c_1 + h.c.$$

We know that 120 degrees rotation leaves the Hamiltonian invariant

What are the eigenvalues under the rotation symmetry operation?

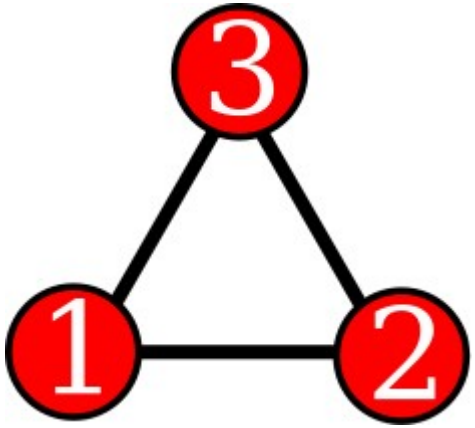
Rotation operator

Symmetry eigenvalue

$$R|\Psi\rangle = e^{i\phi}|\Psi\rangle$$

Guessing the form of wavefunctions

Take this Hamiltonian



$$R|\Psi\rangle = e^{i\phi}|\Psi\rangle$$

Rotating three times brings the system back to itself

$$R^3 = I$$

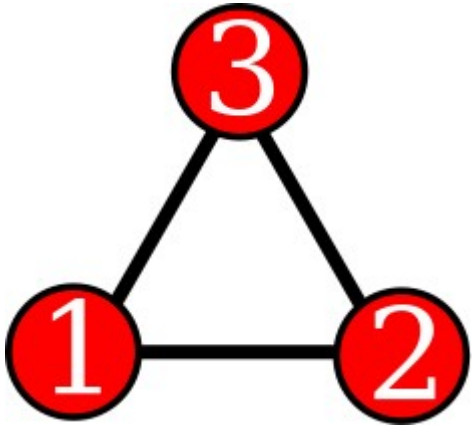
$$(e^{i\phi})^3 = 1$$

Possible phases

$$\phi = \frac{2n\pi}{3} \qquad n = 0, 1, 2$$

Guessing the form of wavefunctions

Take this Hamiltonian



$$\phi = \frac{2n\pi}{3} \quad n = 0, 1, 2$$

$$R|\Psi\rangle = e^{i\phi}|\Psi\rangle$$

The possible form of an eigenstate

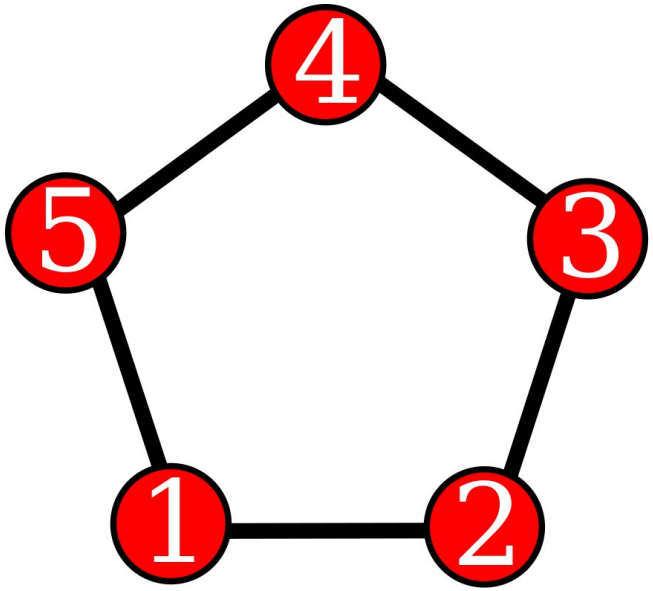
$$\Psi^\dagger = \alpha_1 c_1^\dagger + \alpha_2 c_2^\dagger + \alpha_3 c_3^\dagger$$

α_i Complex number

What are the exact coefficients of the wavefunction?

Guessing a harder wavefunction

Take this Hamiltonian



$$H = \sum_{\langle ij \rangle} c_i^\dagger c_j$$

What are the symmetry eigenvalues?

$$R|\Psi\rangle = e^{i\phi}|\Psi\rangle$$

For the ground state, what is the value of

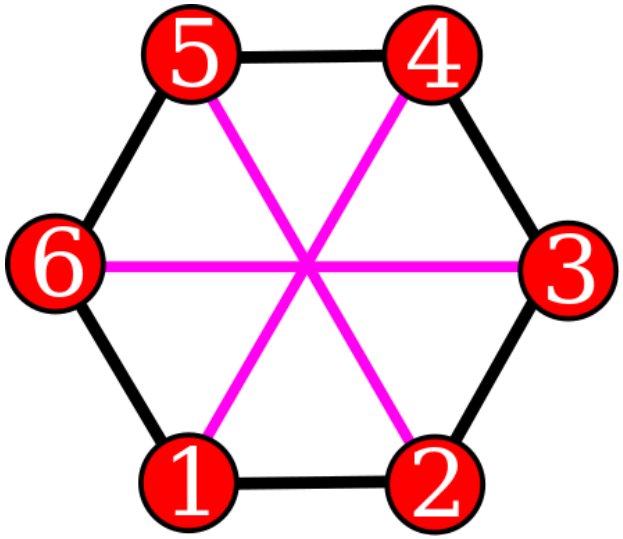
$$|\langle\Omega|c_1 \Psi_{GS}^\dagger|\Omega\rangle|^2$$

vacuum

Ground state

Guessing a harder wavefunction

Take this Hamiltonian



$$H = \sum_{ij} t_{ij} c_i^\dagger c_j$$

What are the symmetry eigenvalues?

$$R|\Psi\rangle = e^{i\phi}|\Psi\rangle$$

For the ground state, what is the value of

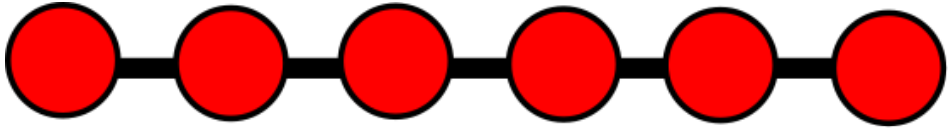
$$|\langle\Omega|c_1 \Psi_{GS}^\dagger|\Omega\rangle|^2$$

vacuum

Ground state

Translational symmetry in chains

One dimensional tight binding chain



$$H = \sum_{n=-\infty}^{\infty} c_n^\dagger c_{n+1} + h.c.$$

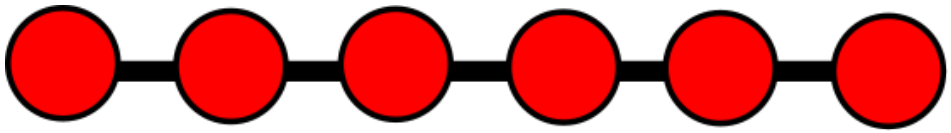
The Hamiltonian commutes with the translation operator → Bloch's theorem

$$T : c_n \rightarrow c_{n+1} \qquad [H, T] = 0 \qquad T|\Psi_\phi\rangle = e^{i\phi}|\Psi_\phi\rangle$$

$\phi \equiv$ Bloch phase of the wavefunction $\qquad \phi \in [0, 2\pi)$

Translational symmetry in chains

One dimensional tight binding chain



$$H = \sum_{n=-\infty}^{\infty} c_n^\dagger c_{n+1} + h.c.$$

$$[H, T] = 0$$

$$T|\Psi_\phi\rangle = e^{i\phi}|\Psi_\phi\rangle$$

$$H|\Psi_\phi\rangle = \epsilon_\phi|\Psi_\phi\rangle$$

The mapping between ϕ and ϵ_ϕ is what we call band structure

$e^{i\phi} \equiv$ Symmetry eigenvalue

$\epsilon_\phi \equiv$ Energy eigenvalue

Translational symmetry in chains

In the original (real-space) basis

$$H = \sum_{n=-\infty}^{\infty} c_n^\dagger c_{n+1} + h.c.$$

In the diagonal basis

$$H = \sum_{\phi} \epsilon_\phi \Psi_\phi^\dagger \Psi_\phi$$

From the symmetry eigenvalue, we can determine the expansion of $\Psi_\phi^\dagger$

$$T\Psi_\phi^\dagger T^{-1} = e^{i\phi}\Psi_\phi^\dagger \qquad \Psi_\phi^\dagger = \sum_n a_{n,\phi} c_n^\dagger \qquad a_{n,\phi} = e^{in\phi}$$

And by taking that eigenfunction, we get the eigenvalues of the Hamiltonian

$$\Psi_\phi^\dagger = \sum_n e^{in\phi} c_n^\dagger \qquad H|\Psi_\phi\rangle = \epsilon_\phi |\Psi_\phi\rangle \qquad \epsilon_\phi = 2\cos\phi$$

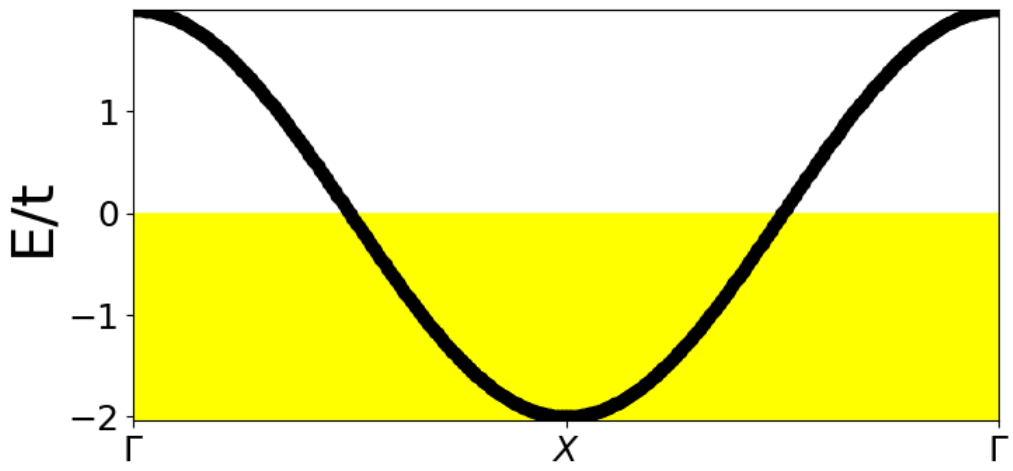
Computing electronic band structures with Python

pyqula Python library

Installation

```
pip install pyqula
```

Band structure

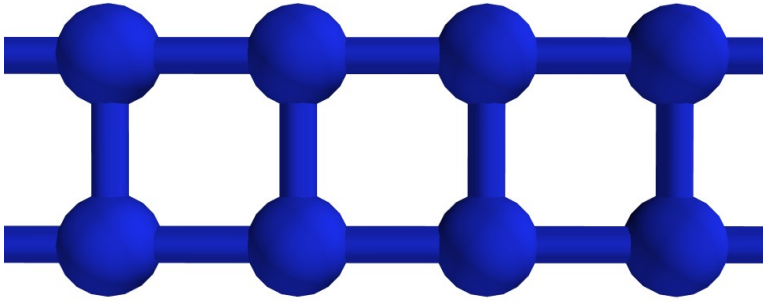


Example script for a 1D tight binding chain

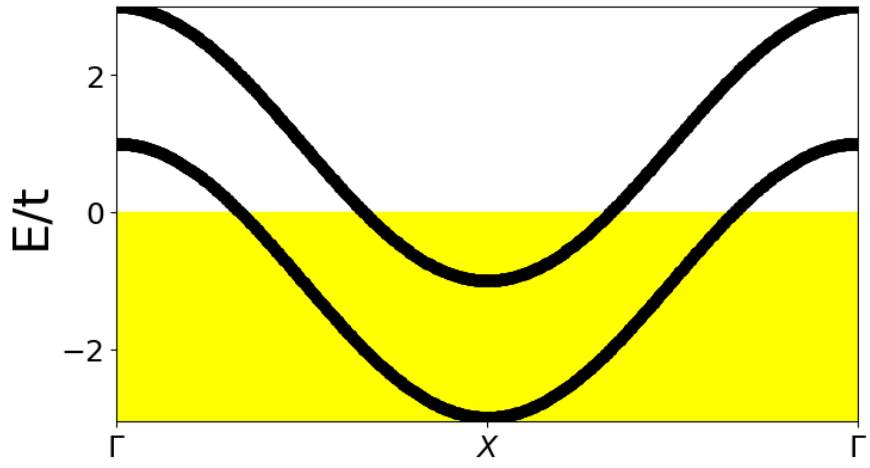
```
from pyqula import geometry
g = geometry.chain() # create a chain geometry
h = g.get_hamiltonian() # get the tight binding Hamiltonian
(k,e) = h.get_bands() # get the band structure
```

Computing electronic band structures with Python

Let us consider now a model in a 1D ladder



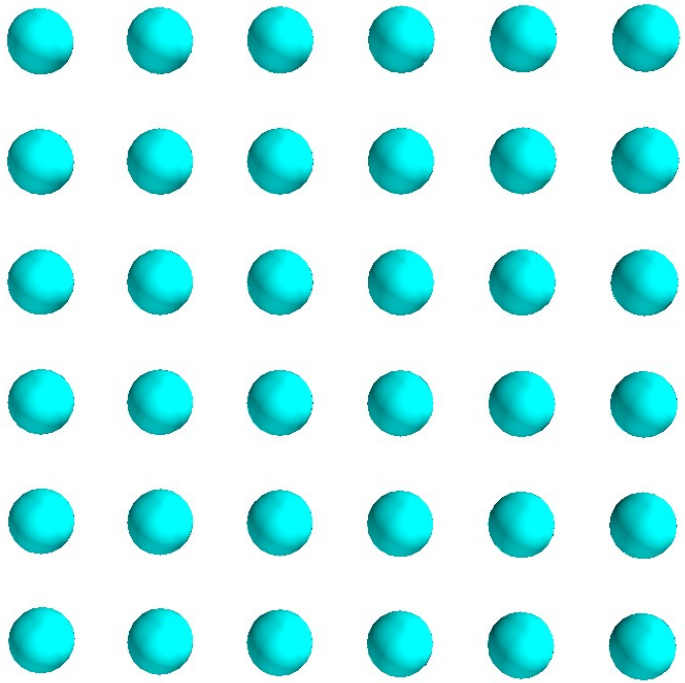
Band structure



Example script for a 1D tight binding ladder

```
from pyqula import geometry
g = geometry.ladder() # create a ladder geometry
h = g.get_hamiltonian() # get the tight binding Hamiltonian
(k,e) = h.get_bands() # get the band structure
```

Translational symmetry in lattices



Translation in y

T_y

T_x Translation in x

Two possible symmetry operations

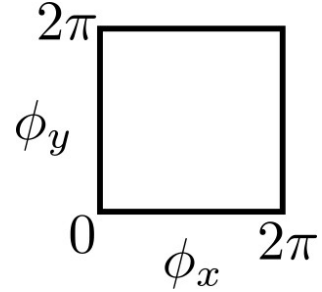
$$T_x|\Psi_{(\phi_x,\phi_y)}\rangle = e^{i\phi_x}|\Psi_{(\phi_x,\phi_y)}\rangle$$

$$T_y|\Psi_{(\phi_x,\phi_y)}\rangle = e^{i\phi_y}|\Psi_{(\phi_x,\phi_y)}\rangle$$

$$\phi_x \in [0, 2\pi) \qquad \phi_y \in [0, 2\pi)$$

The "phases" live in the reciprocal space

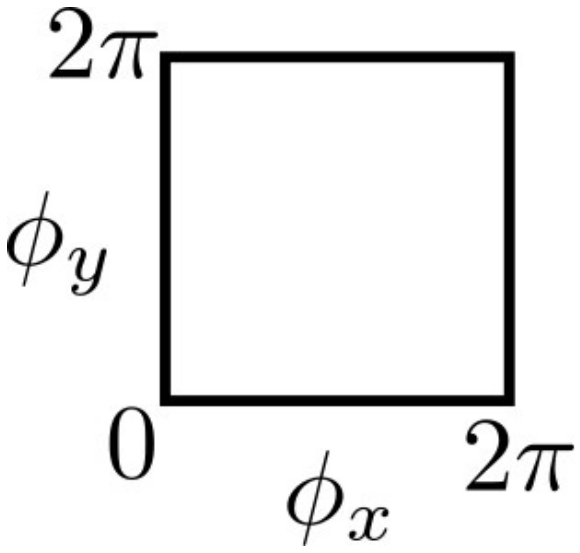
$$\vec{\phi} = (\phi_x, \phi_y) \in$$



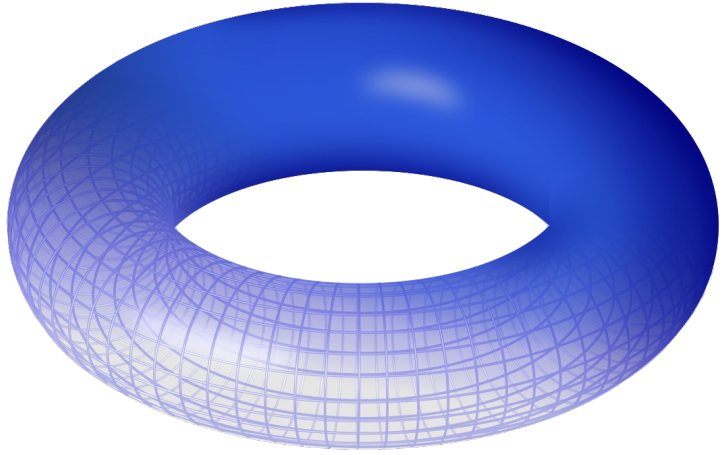
Reciprocal space

The symmetry eigenvalues live in the “reciprocal space”

$$T_x|\Psi_{(\phi_x,\phi_y)}\rangle = e^{i\phi_x}|\Psi_{(\phi_x,\phi_y)}\rangle$$
$$T_y|\Psi_{(\phi_x,\phi_y)}\rangle = e^{i\phi_y}|\Psi_{(\phi_x,\phi_y)}\rangle$$

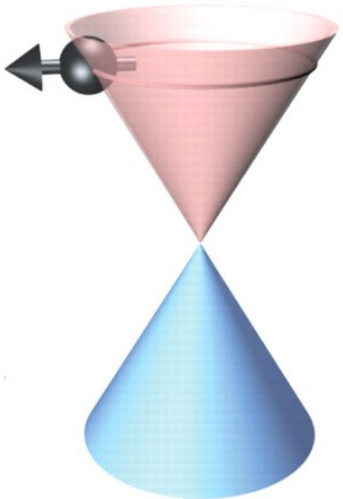


Effectively, the 2D reciprocal space is a torus (periodic boundary conditions)

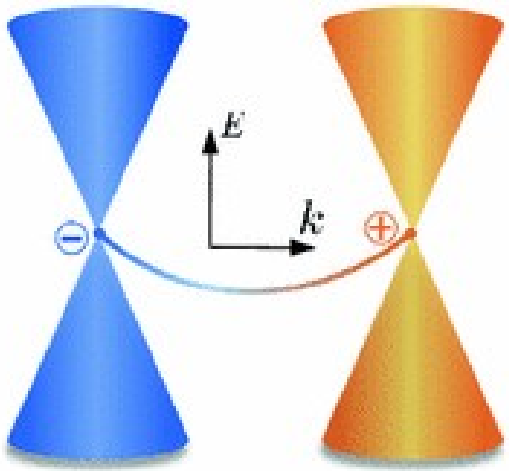


The physics of band-structures

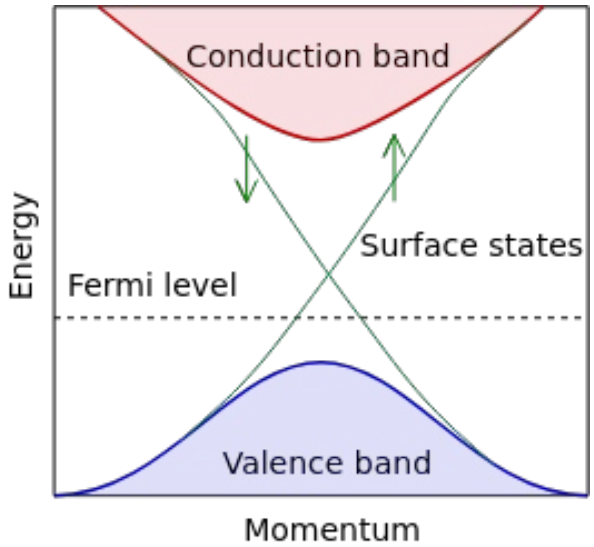
Dirac fermions



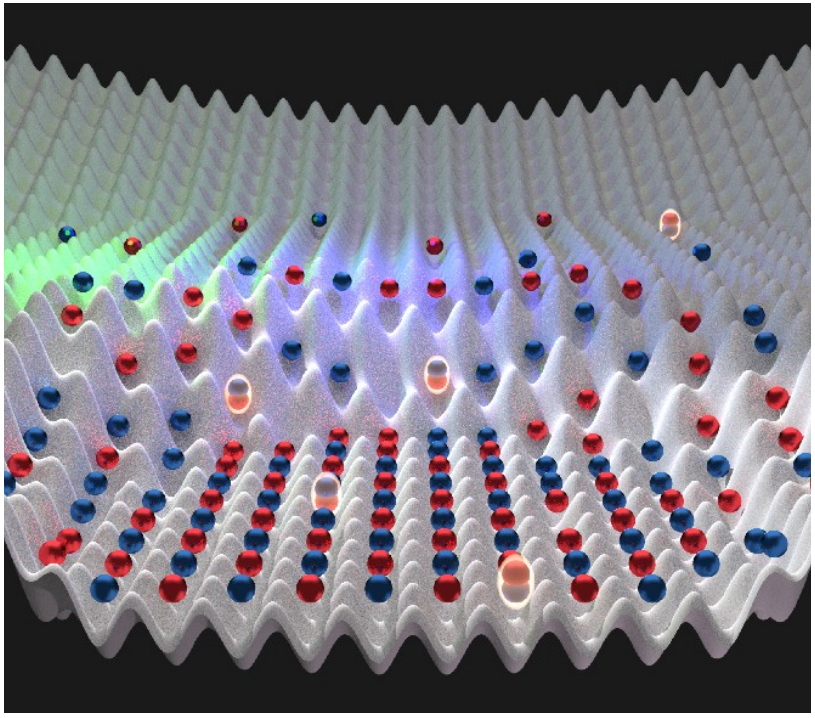
Weyl semimetals

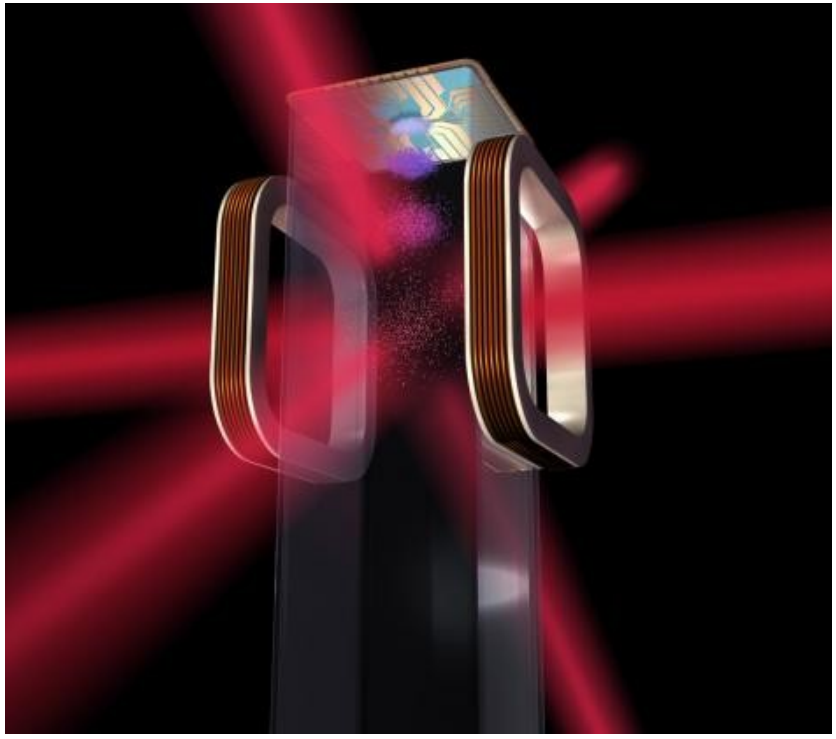


Topological insulators



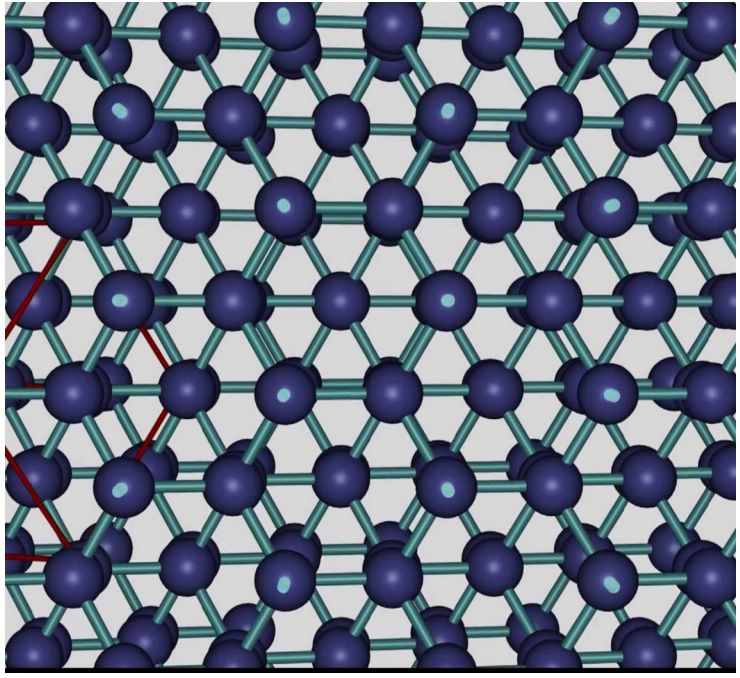
Artificial band-structures with cold-atoms

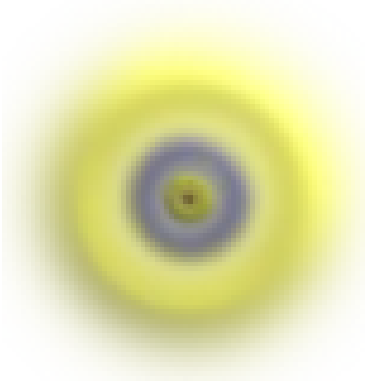




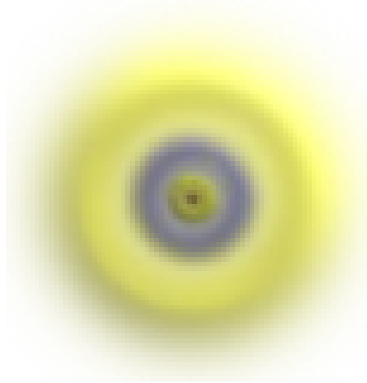
Atoms trapped with lasers allow to realize artificial band structures

Electronic band-structures





Atom #1

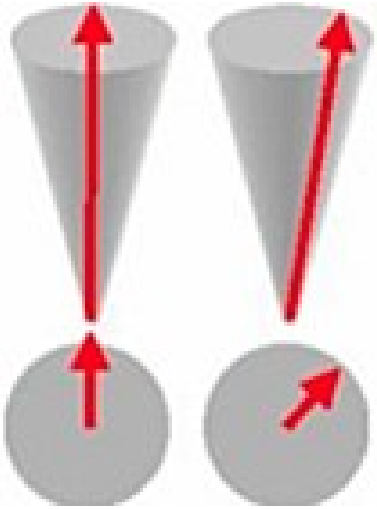


Atom #2

We will focus on the band-structures associated to electronic states

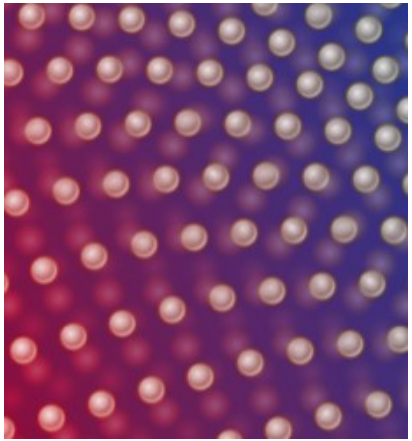
Band-structures in materials beyond electrons

Magnons



Magnets

Phonons



Any crystal

BdG quasiparticles

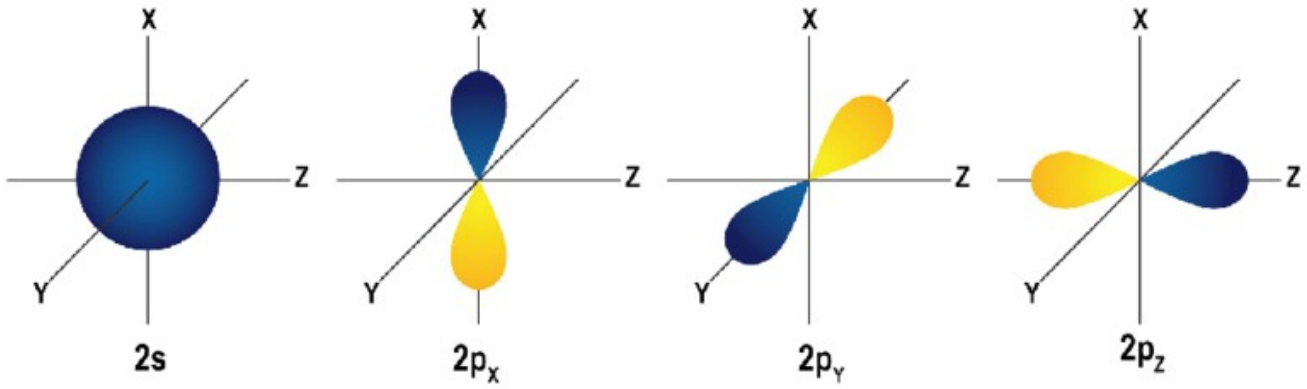


Superconductors

Single orbital band-structures

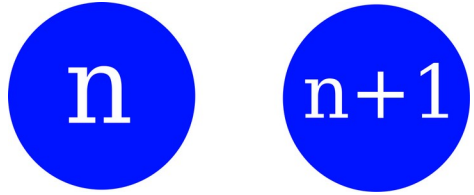
Basis of our Hamiltonian

Atomic orbital (non-orthogonal between different sites)



We will take as basis orthogonalized orbitals (Wannier states)

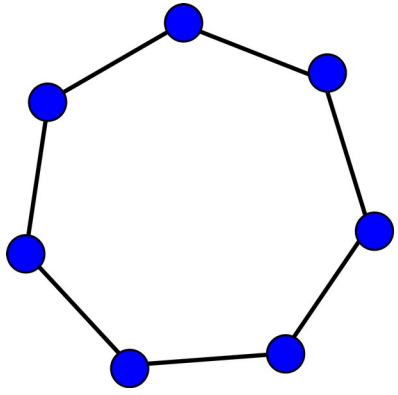
$$\langle n|n+1\rangle = 0 \qquad \int \Psi_n(\mathbf{r})\Psi^*_{n+1}(\mathbf{r})d^N\mathbf{r} = 0$$



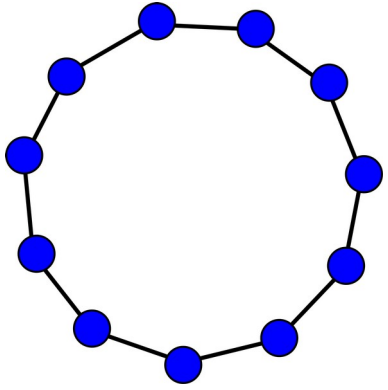
From the finite to the infinite limit

We take a periodic large system, and then the limit when the sites are infinite

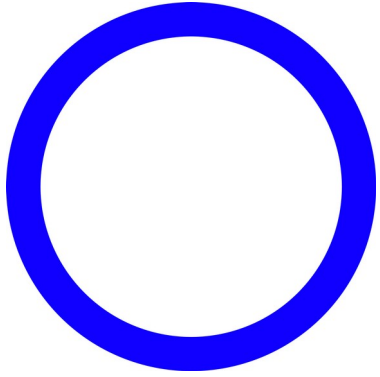
L= 7 sites



L= 11 sites



$L = \infty$ sites



$$\Psi_\phi^\dagger = \frac{1}{\sqrt{L}} \sum_{n=0}^{n=L} e^{in\phi} c_n^\dagger$$

We will take this notation

(for simplicity)

$$\Psi_\phi^\dagger = \sum_n e^{in\phi} c_n^\dagger$$

A note on translational symmetry

For any generic wavefunction

$$\phi(x+a) = e^{i\hat{p}a}\phi(x)$$ ← ***Taylor expansion***

Momentum $\hat{p} = -i\partial_x$

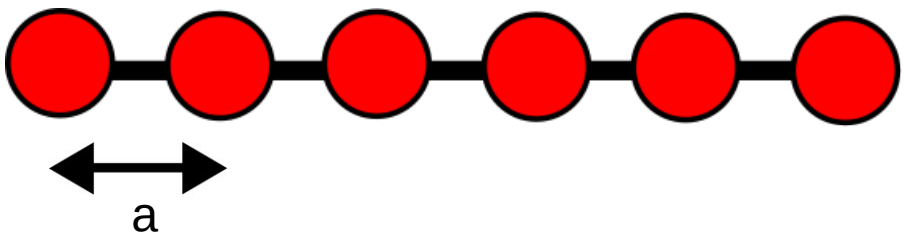
$e^{i\phi}$ symmetry eigenvalue

k Crystal momentum

a lattice constant

For a Bloch wavefunction

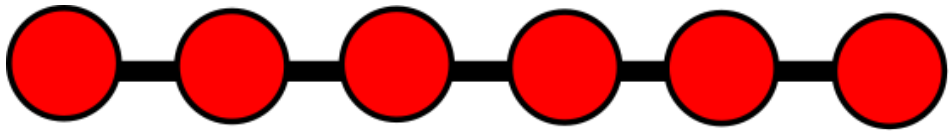
$$\Psi(x+a) = e^{i\phi}\Psi(x)$$



$$e^{i\phi} \equiv e^{ika}$$

One dimensional band structure

We know that for an infinite 1D periodic system



$$H = \sum_n c_n^\dagger c_{n+1} + h.c.$$

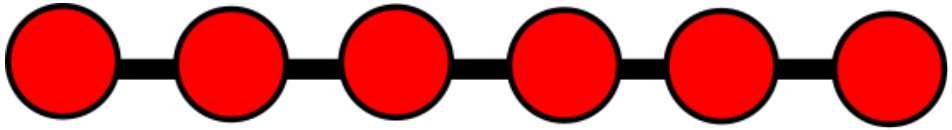
The Hamiltonian is diagonalized as

$$H = \sum_\phi \epsilon_\phi \Psi_\phi^\dagger \Psi_\phi \qquad \Psi_\phi^\dagger \sim \sum_n e^{i\phi n} c_n^\dagger \qquad \langle \Psi_\phi | \Psi_{\phi'} \rangle = \delta_{\phi,\phi'}$$

How do we compute the Hamiltonian eigenvalues ϵ_ϕ?

One dimensional band structure

Hamiltonian 1D periodic system



$$H = \sum_n c_n^\dagger c_{n+1} + h.c.$$

Direct and inverse transformation

$$\Psi_\phi^\dagger \sim \sum_n e^{i\phi n} c_n^\dagger$$

$$c_n^\dagger \sim \sum_\phi e^{-i\phi n} \Psi_\phi^\dagger$$

Plug it in the Hamiltonian

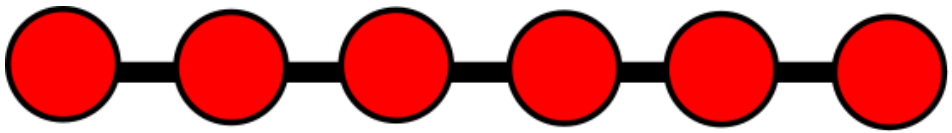
$$\sum_n e^{in(\phi' - \phi)} = \delta_{\phi,\phi'}$$

Orthogonality

$$H = \sum_{n,\phi,\phi'} e^{-i\phi n} \Psi_\phi^\dagger e^{i\phi'(n+1)} \Psi_{\phi'} + h.c. = \sum_\phi e^{i\phi} \Psi_\phi^\dagger \Psi_\phi + h.c. \equiv \sum_\phi H_\phi$$

One dimensional band structure

Hamiltonian 1D periodic system



$$H = \sum_n c_n^\dagger c_{n+1} + h.c.$$

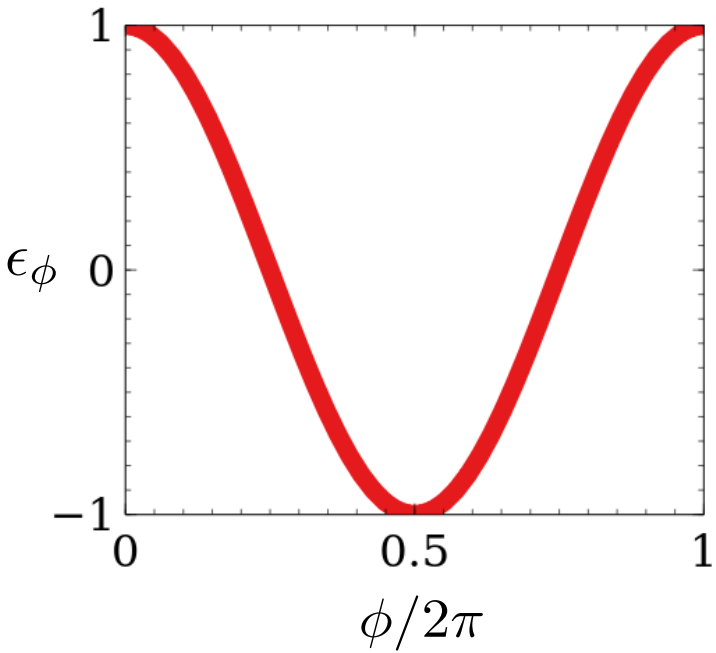
$$H = \sum_\phi \epsilon_\phi \Psi_\phi^\dagger \Psi_\phi$$

$\epsilon_\phi = 2\cos\phi$ ← Electronic dispersion

$\phi \in [0, 2\pi)$

Bloch wavefunctions

$$\Psi_\phi^\dagger \sim \sum_n e^{i\phi n} c_n^\dagger$$



Question: one dimensional band-structure

What is the band-structure of the following Hamiltonian?

$$H = \sum_{i=-\infty}^{\infty} c_i^\dagger c_{i+1} + \eta c_i^\dagger c_{i+2} + h.c.$$

Option A

$$\epsilon_\phi = 2\cos\phi + 2\eta\cos 2\phi$$

Option B

$$\epsilon_\phi = 2\cos\left(\phi + \eta\right)$$

Question: one dimensional band-structure

What is the band-structure of the following Hamiltonian?

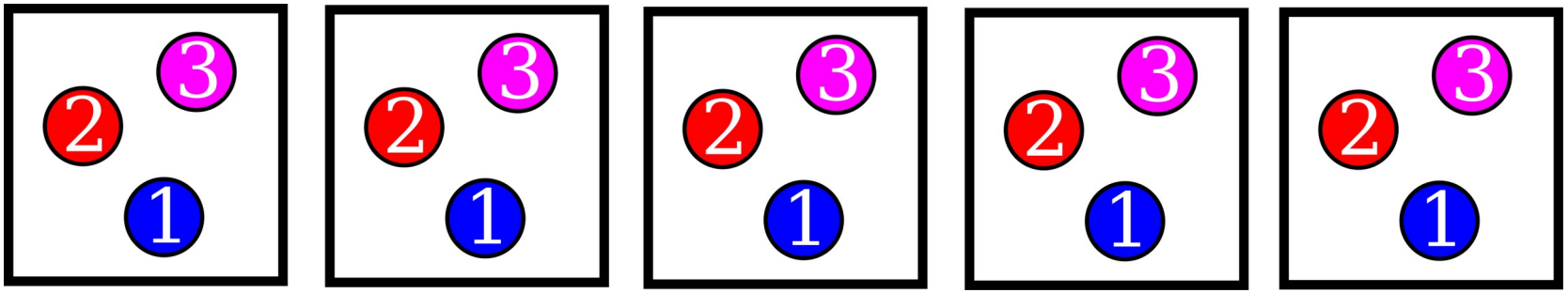
$$H = \sum_{i=-\infty}^{\infty} c_i^\dagger c_{i+1} + \eta c_i^\dagger c_{i+2} + h.c.$$

Solution: **Option A**

$$\epsilon_\phi = 2\cos\phi + 2\eta\cos 2\phi$$

Multi-orbital band-structures

Multi-orbital band-structures

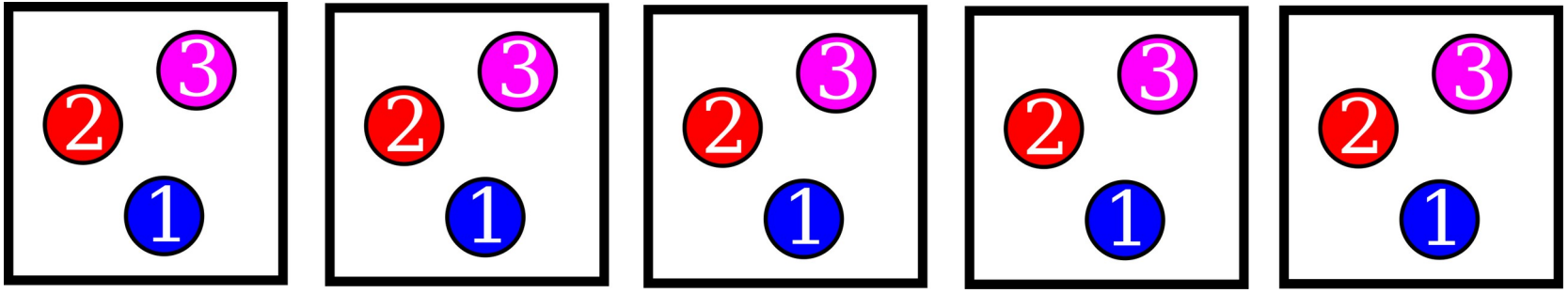


$$c^{\dagger}_{\alpha,n}$$

index of the orbital in the unit cell

Index of the unit cell

Multi-orbital band-structures



$$H = \sum_{n,\alpha,\beta} t_{\alpha\beta} c^{\dagger}_{\alpha,n} c_{\beta,n} + \sum_{n,\alpha,\beta} \gamma_{\alpha\beta} c^{\dagger}_{\alpha,n} c_{\beta,n+1} + h.c.$$

Intra-cell hoppings

Inter-cell hoppings

Multi-orbital band-structures

$$H = \sum_{n,\alpha,\beta} t_{\alpha\beta} c^{\dagger}_{\alpha,n} c_{\beta,n} + \sum_{n,\alpha,\beta} \gamma_{\alpha\beta} c^{\dagger}_{\alpha,n} c_{\beta,n+1} + h.c.$$

Unitary transformation

$$\Psi^{\dagger}_{\phi,\alpha} \sim \sum_{n,\beta} e^{i\phi n} U_{\alpha\beta} c^{\dagger}_{n,\beta} \qquad H = \sum_{\phi,\alpha} \epsilon_{\phi,\alpha} \Psi^{\dagger}_{\phi,\alpha} \Psi_{\phi,\alpha}$$

$\epsilon_{\phi,\alpha}$ **are the eigenvalues of the matrix**

$$h(\phi) = t_{\mu\nu} + \gamma_{\mu\nu} e^{i\phi} + h.c.$$

Question: one dimensional multi-orbital band-structure

What is the band-structure of the following Hamiltonian?

$$H = \sum_{i=-\infty}^{\infty} c^{\dagger}_{A,i} c_{B,i} + \eta c^{\dagger}_{B,i} c_{A,i+1} + h.c.$$

Option A

$$\epsilon_{\phi} = \pm |1 + \eta e^{-i\phi}|$$

Option B

$$\epsilon_{\phi} = 2\cos\phi + 2\eta\cos\phi$$

Question: one dimensional multi-orbital band-structure

What is the band-structure of the following Hamiltonian?

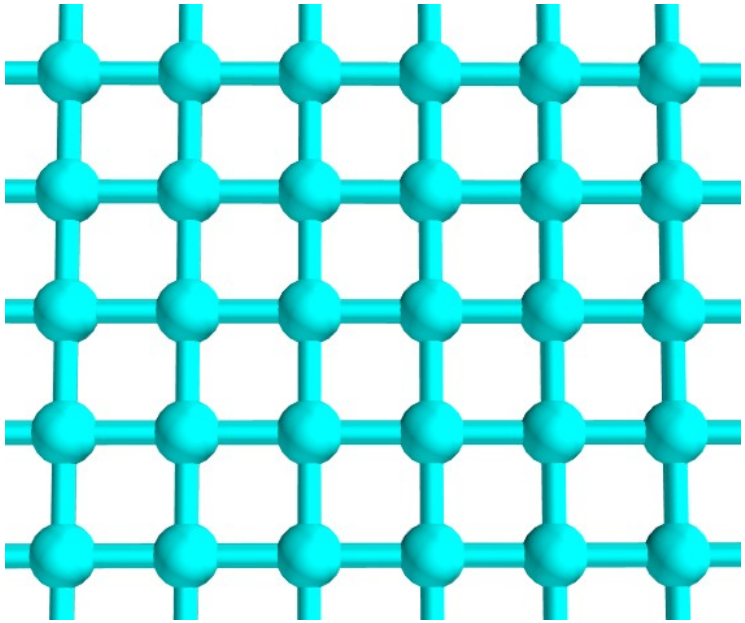
$$H = \sum_{i=-\infty}^{\infty} c^{\dagger}_{A,i} c_{B,i} + \eta c^{\dagger}_{B,i} c_{A,i+1} + h.c.$$

Solution: $h(\phi) = \begin{pmatrix} 0 & 1+\eta e^{-i\phi} \\ 1+\eta e^{i\phi} & 0 \end{pmatrix}$

Option A $\epsilon_{\phi} = \pm|1+\eta e^{-i\phi}|$

Higher dimensional band-structures

Higher dimensional band-structures



T_y Translation in y

T_x Translation in x

Two possible symmetry operations

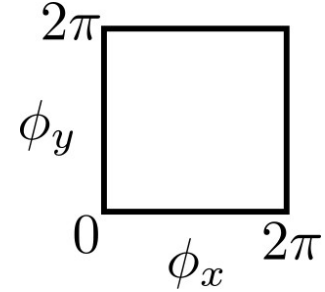
$$T_x|\Psi_{(\phi_x,\phi_y)}\rangle = e^{i\phi_x}|\Psi_{(\phi_x,\phi_y)}\rangle$$

$$T_y|\Psi_{(\phi_x,\phi_y)}\rangle = e^{i\phi_y}|\Psi_{(\phi_x,\phi_y)}\rangle$$

$$\phi_x \in [0, 2\pi) \qquad \phi_y \in [0, 2\pi)$$

The "phases" live in the reciprocal space

$$\vec{\phi} = (\phi_x, \phi_y) \in$$



Reciprocal space

The phase is associated to the Bloch wavevector $\vec{k}$

$$\phi_x \equiv \vec{a}_1 \cdot \vec{k}$$

$$\phi_y \equiv \vec{a}_2 \cdot \vec{k}$$

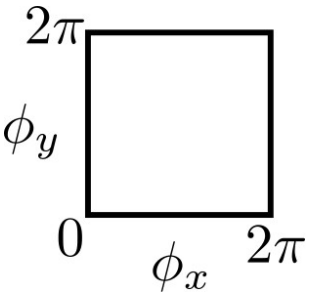
$\vec{k}$ Is a vector in the Brillouin zone

Two possible symmetry operations

$$T_x|\Psi_{(\phi_x,\phi_y)}\rangle = e^{i\phi_x}|\Psi_{(\phi_x,\phi_y)}\rangle$$

$$T_y|\Psi_{(\phi_x,\phi_y)}\rangle = e^{i\phi_y}|\Psi_{(\phi_x,\phi_y)}\rangle$$

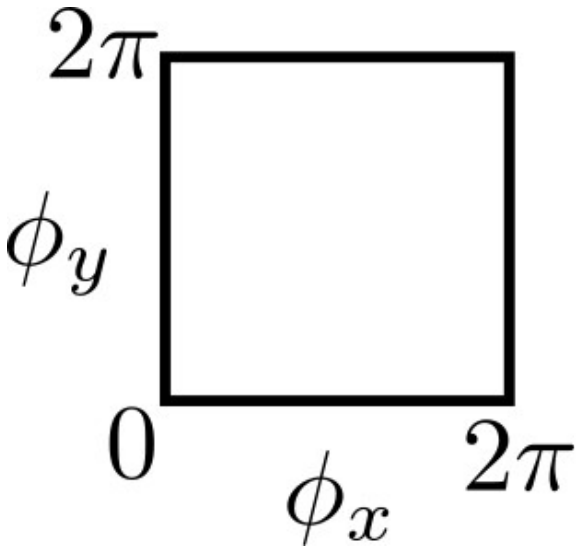
$$\vec{\phi} = (\phi_x, \phi_y) \in$$



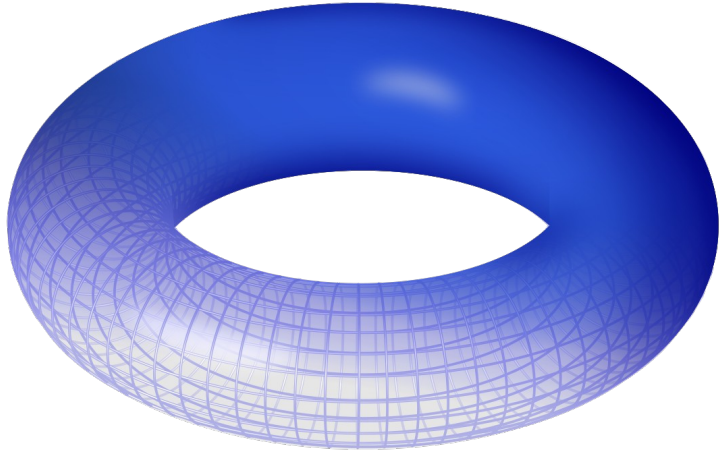
Reciprocal space

The symmetry eigenvalues live in the “reciprocal space”

$$T_x|\Psi_{(\phi_x,\phi_y)}\rangle = e^{i\phi_x}|\Psi_{(\phi_x,\phi_y)}\rangle$$
$$T_y|\Psi_{(\phi_x,\phi_y)}\rangle = e^{i\phi_y}|\Psi_{(\phi_x,\phi_y)}\rangle$$

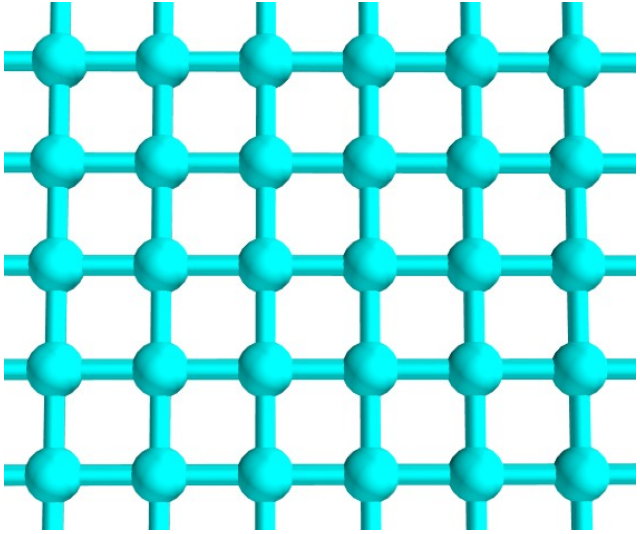


Effectively, the 2D reciprocal space is a torus (periodic boundary conditions)



Exercise: two-dimensional band-structure

What is the band-structure of a single orbital in a square lattice?

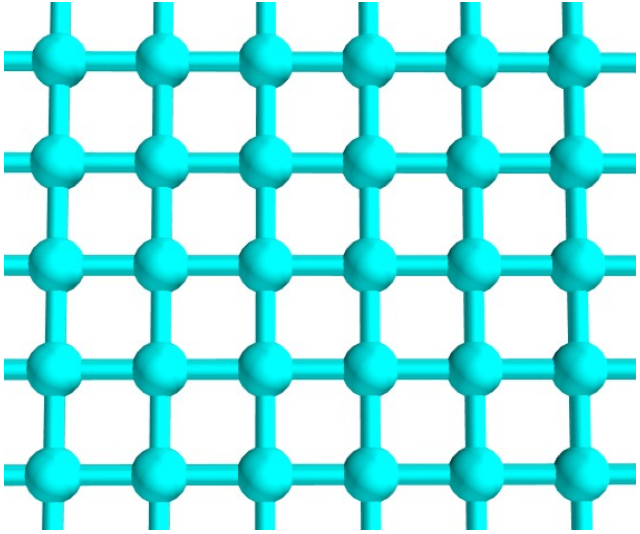


$$H = \sum_{\langle ij \rangle} c_i^\dagger c_j$$

Sum over first neighbors

Exercise: two-dimensional band-structure

What is the band-structure of a single orbital in a square lattice?

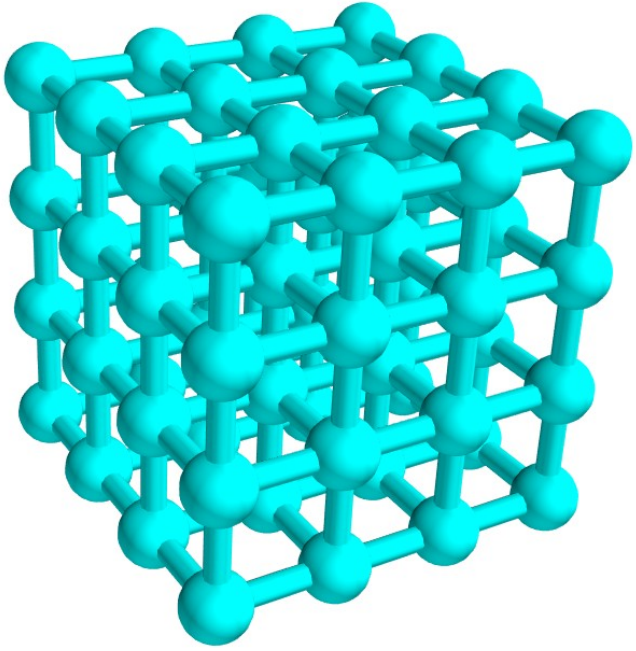


$$H = \sum_{\langle ij \rangle} c_i^\dagger c_j$$

$$\epsilon(k) = 2\cos k_x + 2\cos k_y$$

Exercise: three-dimensional band-structure

What is the band-structure of a single orbital in a cubic lattice?

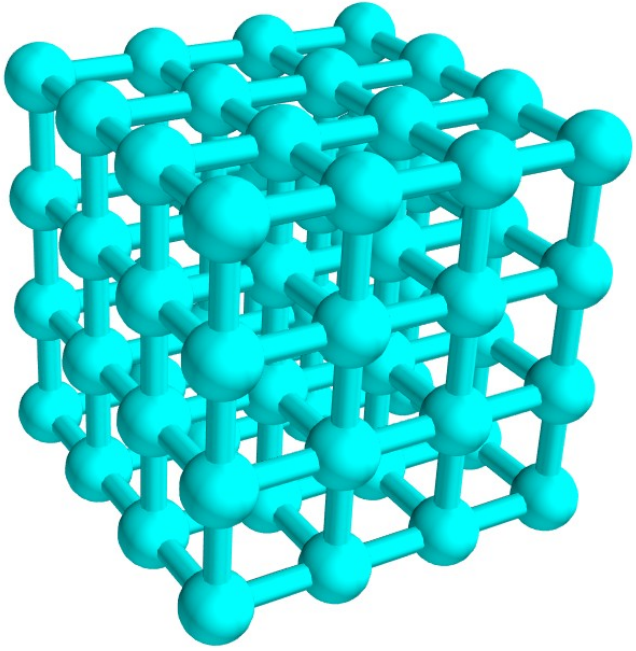


$$H = \sum_{\langle ij \rangle} c_i^\dagger c_j$$

Sum over first neighbors

Exercise: three-dimensional band-structure

What is the band-structure of a single orbital in a cubic lattice?



$$H = \sum_{\langle ij \rangle} c_i^\dagger c_j$$

$$\epsilon(k) = 2\cos k_x + 2\cos k_y + 2\cos k_z$$

Macroscopic properties

Properties of the electronic dispersion

From now on, lets work with a specific electronic dispersion $\epsilon_{\vec{k}}$

Density of states $$D(\omega) \sim \int \delta(\omega - \epsilon_{\vec{k}}) d^N \vec{k}$$

Group velocity

$$v_F = \frac{\partial \epsilon_{\vec{k}}}{\partial k_\alpha}$$

Effective mass

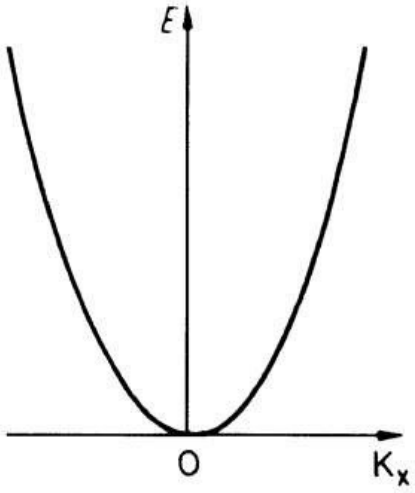
$$\frac{1}{m_{\alpha\beta}} = \frac{\partial^2 \epsilon_{\vec{k}}}{\partial k_\alpha \partial k_\beta}$$

Fermi surface

$$\{\vec{k}\} \text{ with } \epsilon_{\vec{k}} = \epsilon_F$$

Three important electronic dispersions

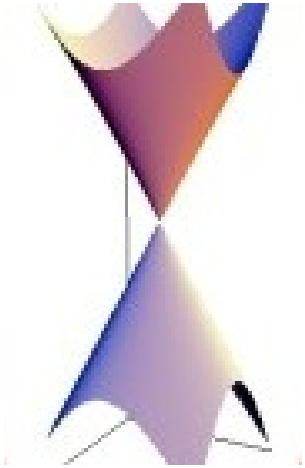
Parabolic bands



Semiconductors, metals

Effective free-electrons

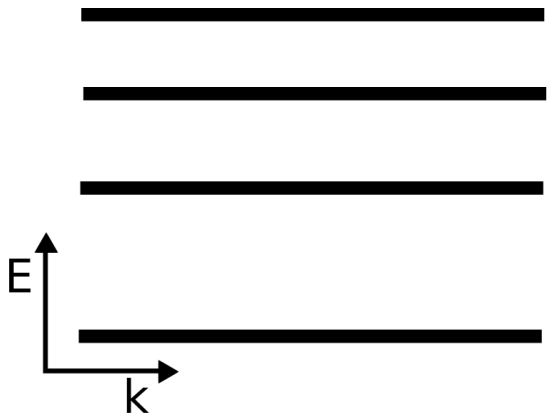
Dirac dispersion



graphene

Topology & relativistic physics

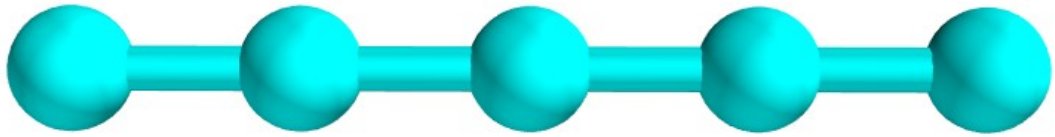
Flat bands



Quantum Hall

Topology & correlations

Parabolic dispersion



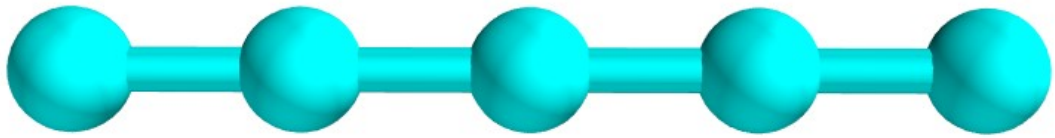
Small k expansion

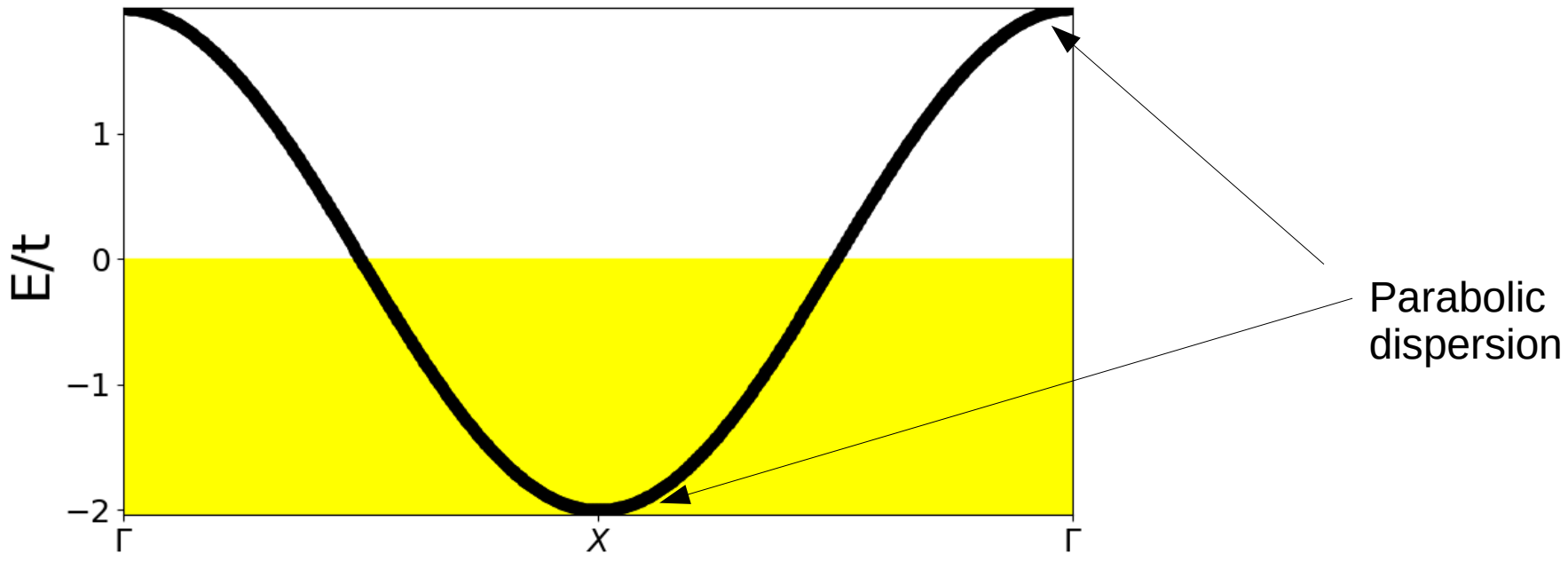
$$\epsilon(k) = 2t\cos ka \approx 2t\left[1 - \frac{k^2a^2}{2}\right]$$

$$\epsilon(k) \sim k^2$$

Conventional parabolic dispersion

Parabolic dispersion





Solving tight binding models

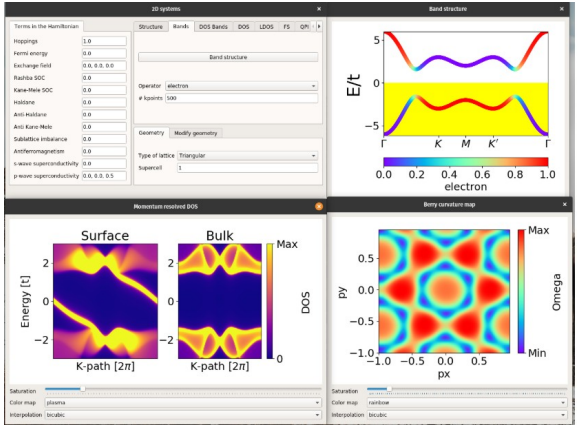
pyqula

```
from pyqula import geometry
g = geometry.honeycomb_lattice()
h = g.get_hamiltonian()
h.add_rashba(0.2) # Rashba spin-orbit coupling
h.add_zeeman([0.,0.,0.6]) # Zeeman field
from pyqula import topology
(kx,ky,omega) = h.get_berry_curvature() # compute Berry curvature
c = h.get_chern() # compute the Chern number
```

- Python library
- Ideal for complex models/calculations
- For writing in Python

https://github.com/joselado/pyqula

Quantum-lattice

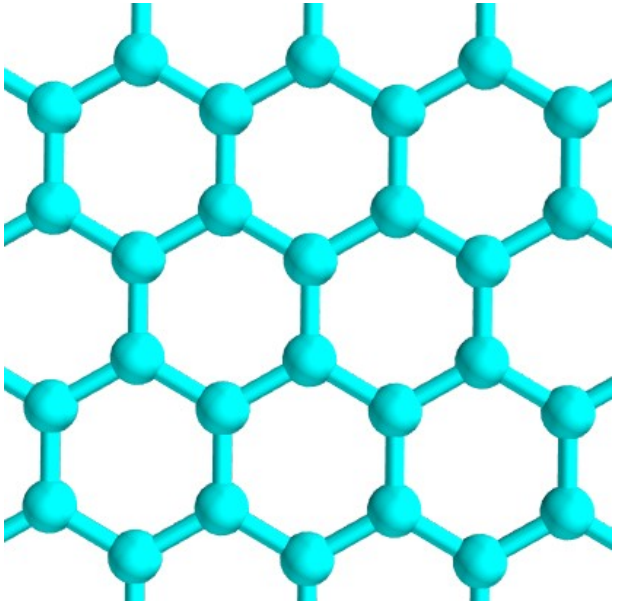


- User-friendly interface for tight binding models
- Ideal for simple models and quick checks
- Fully interface-based, no scripting

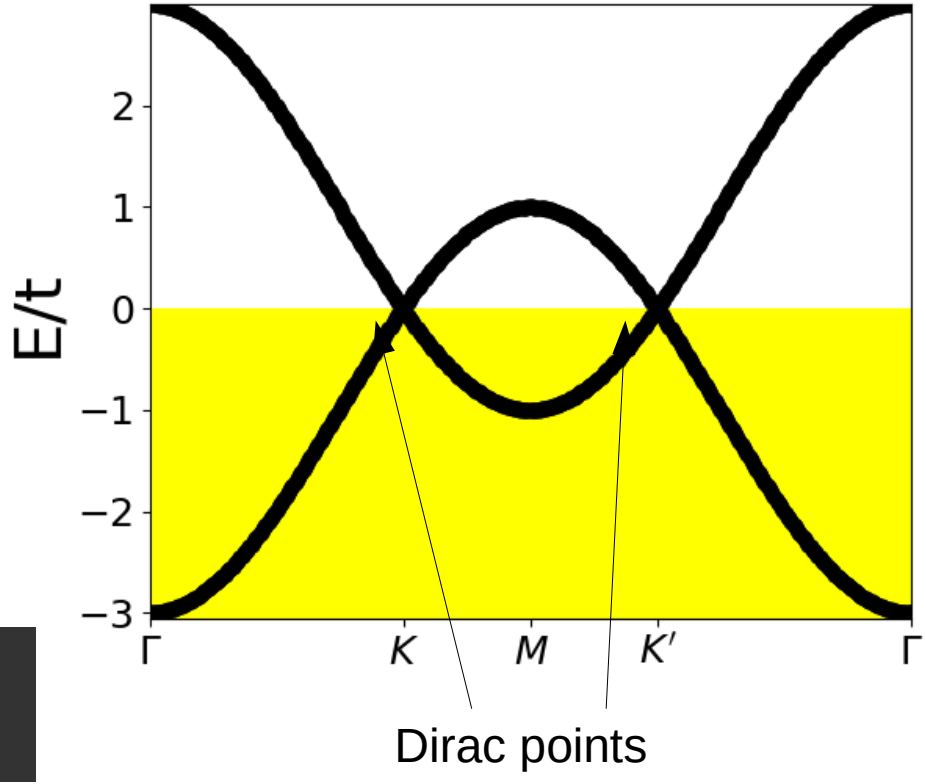
https://github.com/joselado/quantum-lattice

Dirac dispersion

Honeycomb lattice



```
from pyqula import geometry
g = geometry.honeycomb_lattice() # create the lattice
h = g.get_hamiltonian() # get the tight binding Hamiltonian
(k,e) = h.get_bands() # get the band structure
```



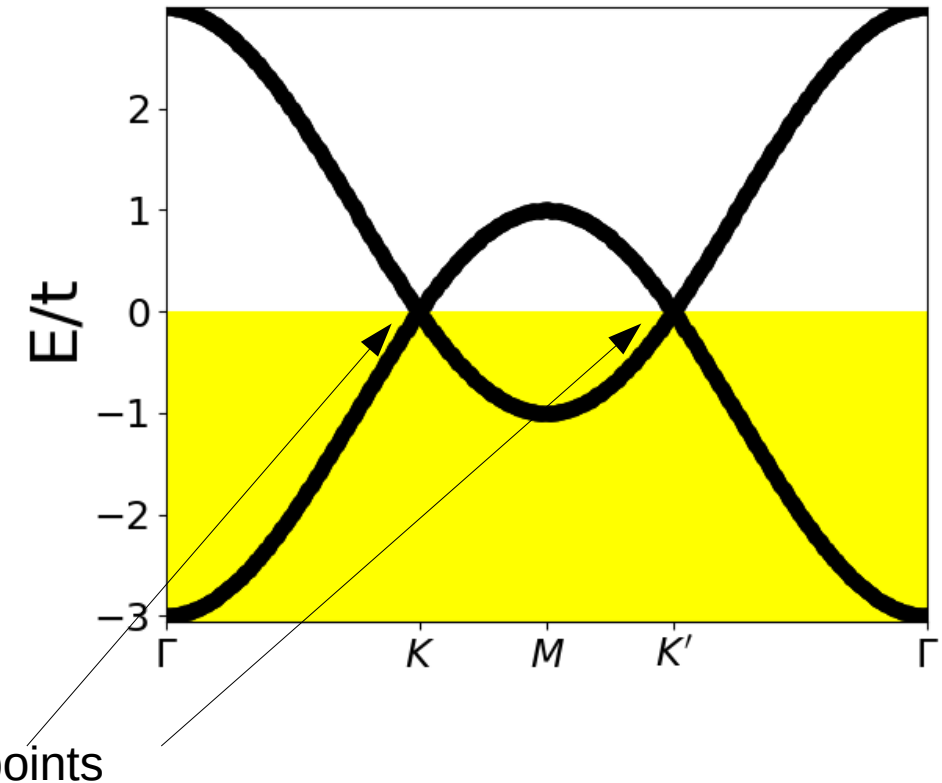
Dirac dispersion

Low energy Hamiltonian at K-points

$$H(k) = \begin{pmatrix} 0 & k_x + ik_y \\ k_x - ik_y & 0 \end{pmatrix}$$

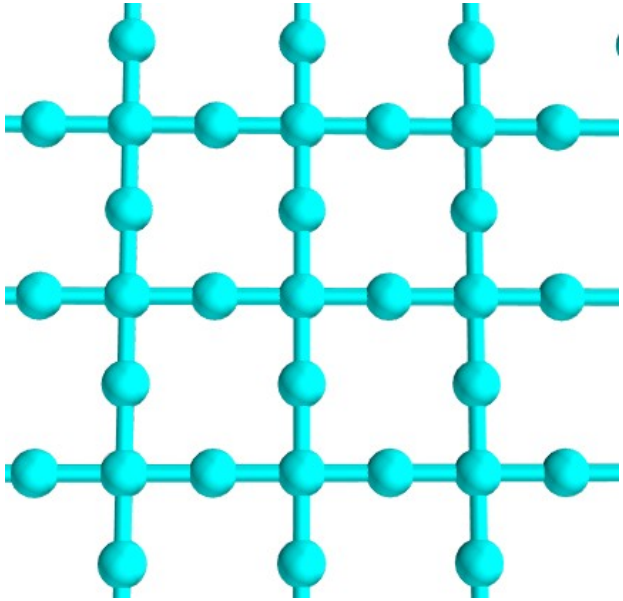
Two-dimensional Dirac equation

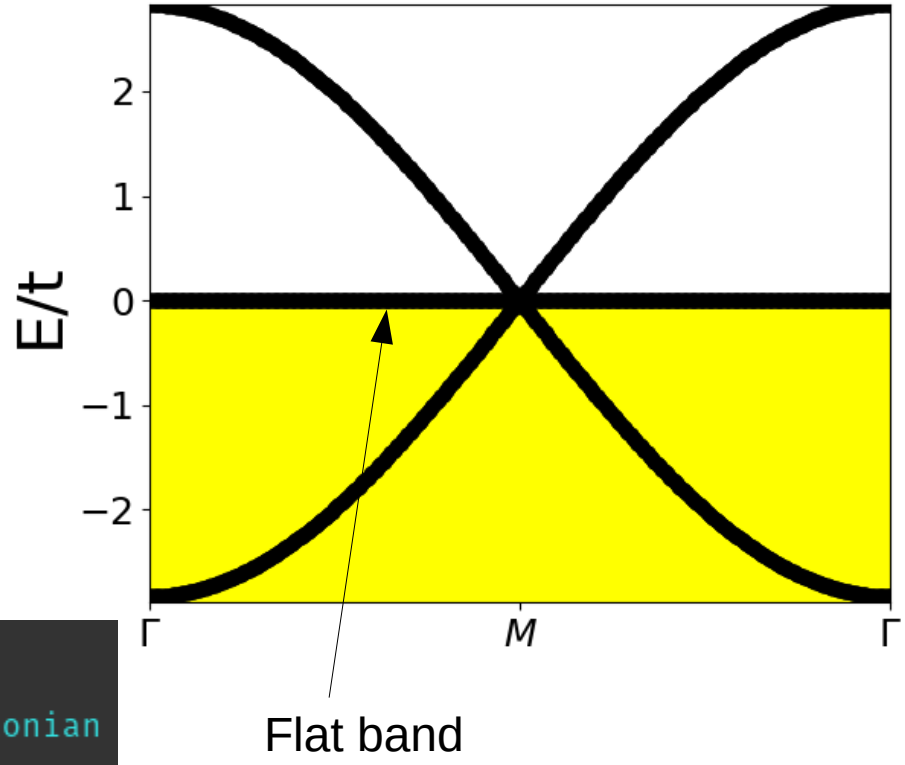
$$\epsilon(k) = \pm\sqrt{k_x^2 + k_y^2}$$



Flat bands

Lieb lattice

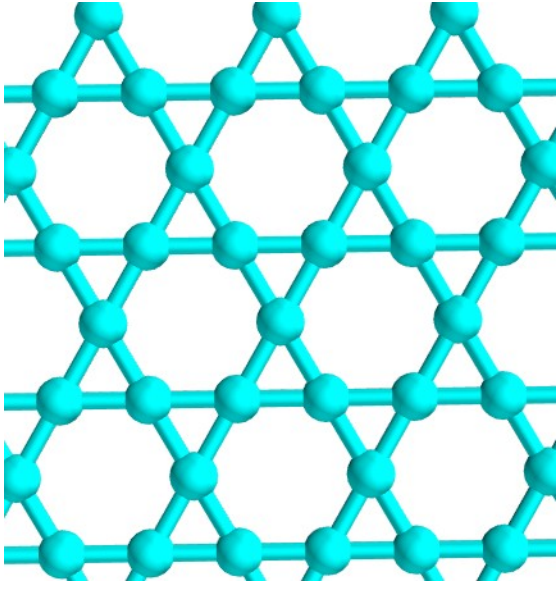




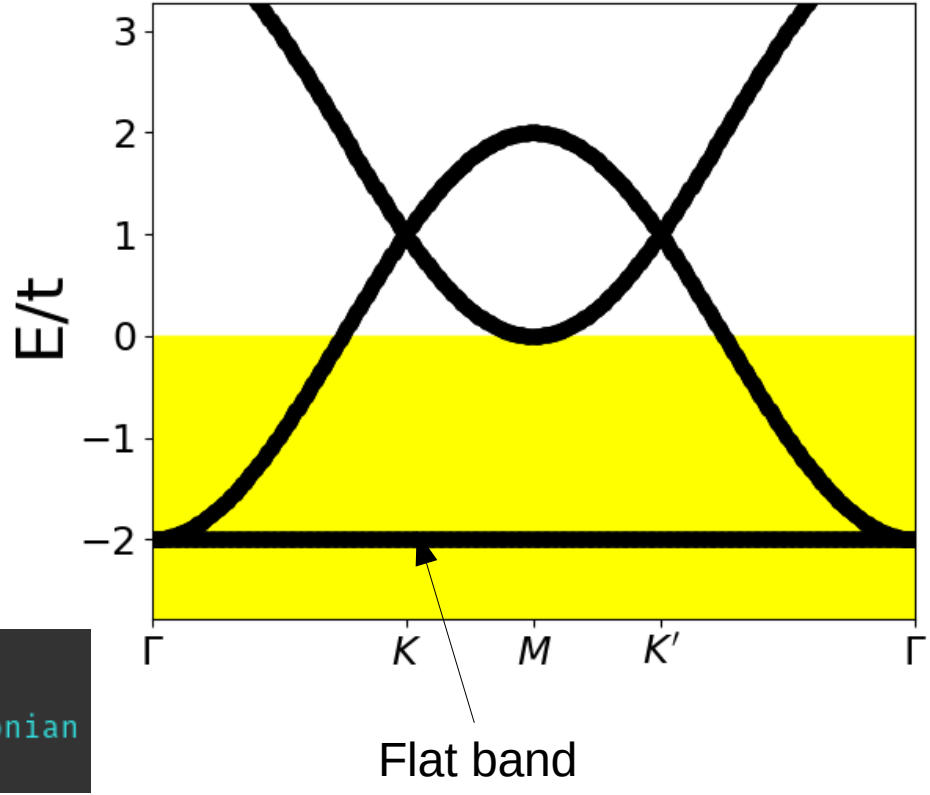

```
from pyqula import geometry
g = geometry.lieb_lattice() # create the lattice
h = g.get_hamiltonian() # get the tight binding Hamiltonian
(k,e) = h.get_bands() # get the band structure
```

Flat bands

Kagome lattice

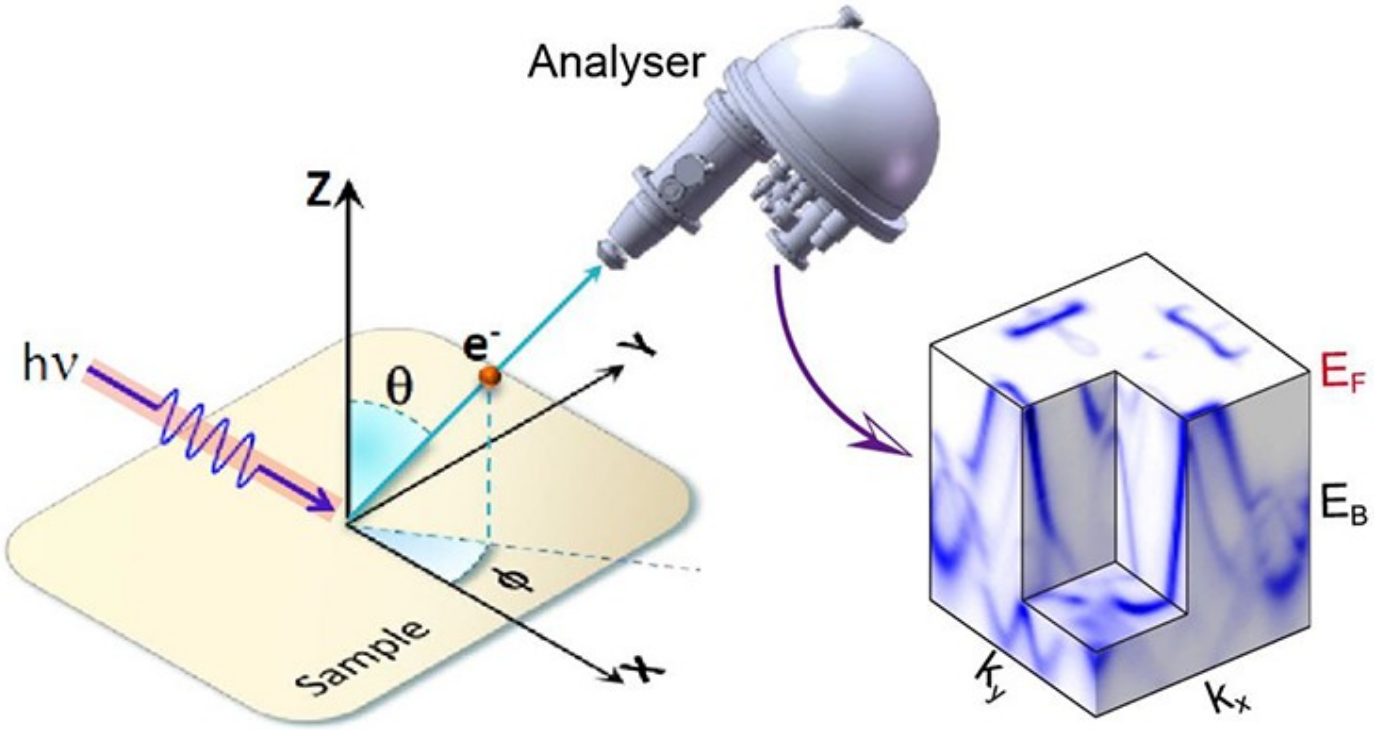


```
from pyqula import geometry
g = geometry.kagome_lattice() # create the lattice
h = g.get_hamiltonian() # get the tight binding Hamiltonian
(k,e) = h.get_bands() # get the band structure
```



Measuring band structures

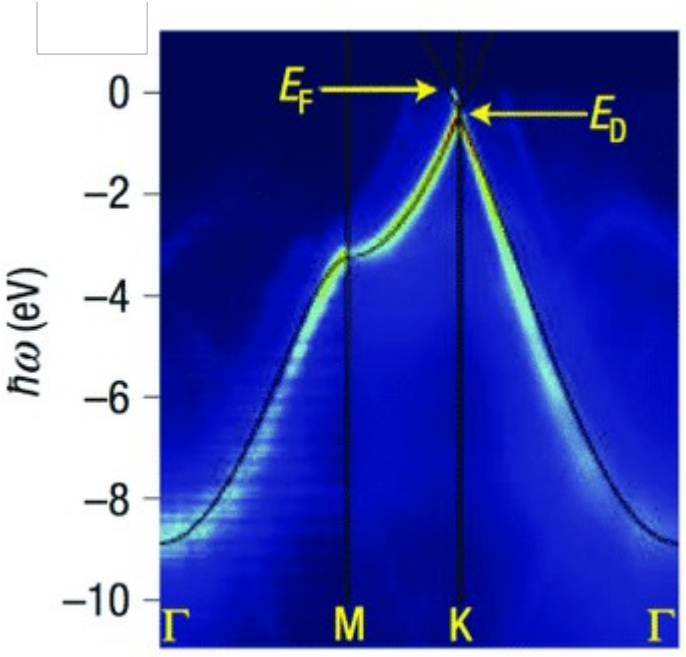
Angle-resolved photoemission spectroscopy



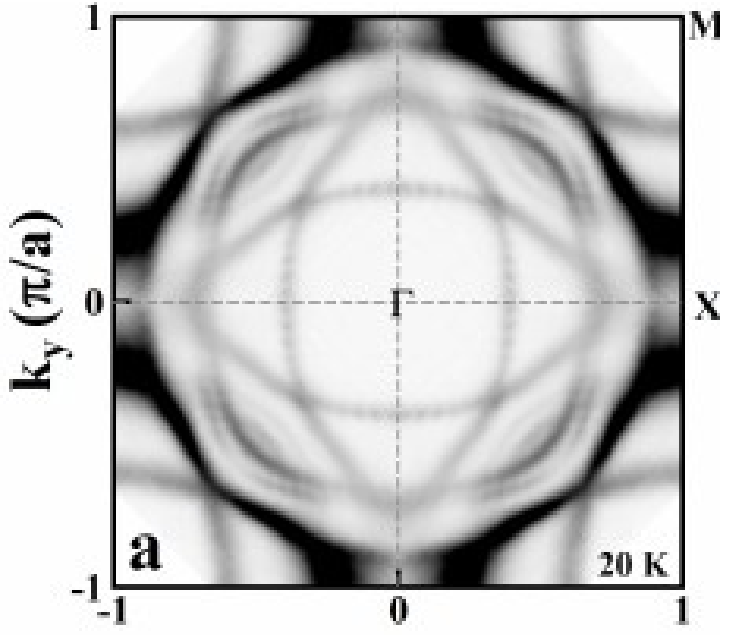
Angle-resolved photoemission spectroscopy (ARPES)

Angle-resolved photoemission spectroscopy

Measuring band-structures with ARPES



Measuring Fermi surfaces with ARPES



Take home

- The spectra of periodic system can be computed with band-structure theory
- The electronic dispersion determines the electronic properties of a compound